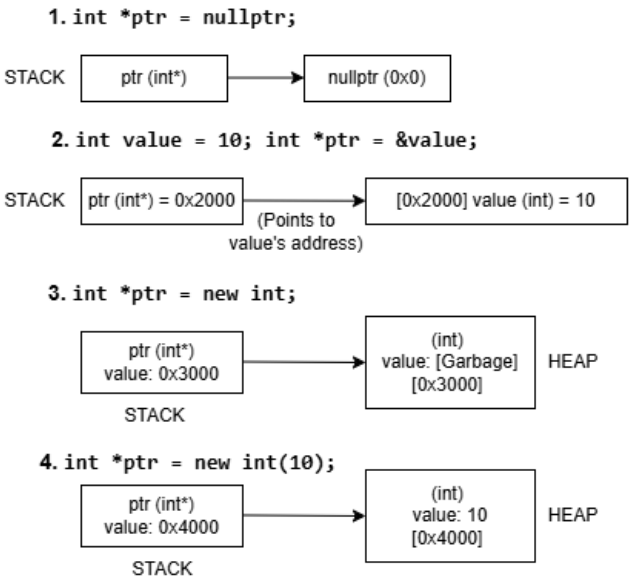
Draw the memory diagram for the following:

```
1. int *ptr = nullptr;
2. int value = 10;
   int  *ptr= &value;
3. int *ptr = new int;
4. int *ptr = new int(10);
```



Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Operator.

Answer: The constructor initializes a new object created in the code to a valid initial state, whereas a destructor is responsible for doing exactly the opposite – that is, clearing everything up just before the destruction of the object. In case of copying, the copy constructor is called in order to create a completely new object in such a way that its values are equal to those of another object, whereas in case of copy assignment, we use another operator in order to change the value of the existing object to be equal to the values of another object.